Pranay Dogra

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Education

University of California, Berkeley | GPA: 4.0 — Berkeley, CA
B.S. in Electrical Engineering & Computer Science and Industrial Engineering & Operations Research — *Grad May 2029*

Experience

Software Engineer — August 2026 – Present
Space Technologies and Rocketry (STAR), UC Berkeley — *Berkeley, CA*
- Developed STM32 flight-computer self-test firmware in C for five onboard components, with SPI diagnostics, bounded retries, and heartbeat scheduling.
- Implemented device-specific checks for the IMU, high-g accelerometer, barometer, Ethernet, and radio, including identity-register reads, calibration PROM CRC validation, and radio error-status inspection.
- Implemented C++ DAQ parsing for temperature, humidity, and pressure telemetry over UDP; integrated board configuration, streaming, and frontend monitoring.
- Decoupled device I/O through callbacks and maintained fixed-size state; validated firmware with CMake/CTest host tests and GitHub Actions CI running AddressSanitizer and UndefinedBehaviorSanitizer.

Software Engineering Intern — August 2026 – Present
Topit AI, LLC | B2B Product
- Built a FastAPI, React, TypeScript, PostgreSQL/pgvector, and Docker platform for personalized content; developed Python and Node.js middleware and feed, CSV, Shopify, WooCommerce, and REST catalog adapters.
- Developed a zero-dependency TypeScript SDK with 250 ms timeouts, decision caching, and synchronous fallbacks; processed 18 real catalog images through ingestion, CLIP embedding, scoring, and API delivery with zero fetch failures.
- Tested API, SDK, and browser workflows across three simulated partners using pytest, Hypothesis, Vitest, and Playwright; used property-based tests, mutation tests, and OpenAPI drift checks.

Founder — November 2024 – August 2026
CueLens | Featured in MidPeninsula Post
- Developed assistive AI for facial-emotion interpretation; partnered with three autism clinics across Taiwan, India, and the U.S., conducted 50+ clinician/user interviews, and raised $5,000 for the Soham Learning Foundation.

Software Engineering Intern — Summers 2024 and 2025
Mirro.ai
- Curated a balanced dataset of 40,000+ videos and images to address emotion-recognition bias; integrated the DeepSruthi model into robotics applications and investor demos.

Research

Undergraduate Researcher — September 2026 – Present
UC Berkeley Haas School of Business, Dana Carney Lab — *Berkeley, CA*
- Develop Python/PyTorch pipelines to extract facial, vocal, and paralinguistic features from audiovisual data for research on political ideology, race, and socioeconomic status.

Student Researcher — June 2023 – August 2023
UC Berkeley AI Research — *Berkeley, CA*
- Built a 1,020-prompt benchmark for bias in agentic LLM decisions; analyzed FLAN, Vicuna, LLaMA, and LLaMA 2 choices/logits. Published *Quantifying Bias in Agentic Large Language Models: A Benchmarking Approach* at IEEE ICTC 2024.

Projects

HANDOFF | TypeScript, MCP, PostgreSQL, Ollama, WebAssembly — GitHub
- Distributed local-model inference across peer machines with independent judge verification; project benchmarks reported 30.8% faster task completion. Built a separate quality, latency, and token-cost evaluation harness.
- Coordinated workers using atomic claims, durable task states, and idempotent request IDs to recover from disconnects; sandboxed workflows in Wassette/Wasmtime and tested host-access restrictions and competing claims.

Babel | LLMs, Finite State Machines, Multi-Agent Systems — GitHub
- Improved structured finite-state-machine generation reliability from 36.9% to 95.3% through multi-agent validation feedback and GBrain shared memory; supported deterministic validation and scenario simulation.

SLM Cascade | Ollama, OpenAI, Claude — GitHub
- Reduced token cost by 70%+ with 3.43-second median latency in a project benchmark using small-model-first routing, local planning, and validation.

Technical Skills

Languages: C++, C, Python, Java, TypeScript, JavaScript, SQL
Embedded & Testing: STM32, SPI, UDP, CMake/CTest, pytest, Hypothesis, GitHub Actions, Vitest, Playwright
Tools & Data: Git, Docker, Kubernetes, PyTorch, NumPy, scikit-learn, FastAPI, PostgreSQL, Node.js